



The Soil — FULL MIND MAP

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Chapter 45_The_Soil

Topic Index

1. Agricultural Practices — Sahi Order
2. Water-Holding Capacity — Soil Types
3. Soil Conservation — Definition
4. Terra Rossa — Limestone se Development
5. Soil Leaching — Tropical Rainforests
6. Halophytes — Saline Soils
7. Soil mein Nitrogen — Sources
8. Clay — Water Holding aur Pore Size
9. Contour Bunding — Mountain Regions
10. Soil Types — Climate Region Matching
11. Earthworms — Soil ke liye Beneficial
12. Agricultural Soils — Organic Matter, Sulphur Cycle, Salinization
13. Edaphic Factor — Soil se Related
14. Geological Formations — Chronological Order
15. Geographical Cycle of Erosion — W.M. Davis

Agricultural Practices — Sahi Order

- Correct order: **Soil Preparation** → **Sowing** → **Irrigation** → **Harvesting**. [Q1]

Water-Holding Capacity — Soil Types

- Water-holding capacity ka decreasing order: **Clay** > **Silt** > **Sand**. [Q2]
- Capillary activity sabse effective **Clayey soil** mein hoti hai (chhote pores) — sabse kam Sandy soil mein (bade pores). Order: **Clay** > **Silt** > **Loamy** > **Sandy**. [Q3]

Soil Conservation — Definition

- Soil conservation = **mitti ko harm se bachana** — broad sense mein sirf erosion control nahi, land ki high fertility maintain karna bhi شامل hai (land-use improvement). [Q4]

Terra Rossa — Limestone se Development

- **Terra Rossa** = red clay soil hai jo **Limestone aur Dolomite** rocks ke weathering se banti hai. [Q5]

Soil Leaching — Tropical Rainforests

- Soil leaching ki major problem **Tropical Rainforests** mein hoti hai — saal bhar heavy rainfall ki wajah se. [Q6]

Halophytes — Saline Soils

- **Halophytes saline soil** mein achhe se ugte hain — Cryophytes (cold soil), Lithophytes (rocks), Xerophytes (tropical desert), Hygrophytes (waterlogged), Hydrophytes (aquatic) alag categories hain. [Q7]

Soil mein Nitrogen — Sources

- Soil mein nitrogen add karte hain: **Animals ka urea excretion AUR vegetation ki death** (decomposers inhe decompose karke nutrients release karte hain, nitrogen cycle chalu rehta hai) — **coal burning NAHI** add karta hai. [Q8]

Clay — Water Holding aur Pore Size

- Clay soil ki water-holding capacity sabse zyada hai (A **TRUE**, particles extremely small, large surface area) — **clay mein large pores NAHI** hote (R **FALSE**, actually bahut fine micropores hote hain). [Q9]

⚠ EXAM TRAP

- Clay mein pores **bade NAHI, bahut fine/micro** hote hain — high water-holding capacity particle size (small) aur surface area (large) ki wajah se hai, pore size ki wajah se NAHI. [Q9]

Contour Bunding — Mountain Regions

- **Contour bunding** soil conservation method **mountain regions** mein use hoti hai (given options — desert margins, flood-prone plains, scrublands — sab galat hain) — rainwater flow slow karke soil erosion check karti hai. [Q10]

Soil Types — Climate Region Matching

- Podzol-Cold temperate, Chernozem-Temperate cold steppe, Spodosols-Humid cold temperate, Laterite-Hot and humid. [Q11]

Earthworms — Soil ke liye Beneficial

- Earthworms agriculture ke liye achhe NAHI hote — yeh statement **galat** hai (A **FALSE**) — actually 'ecosystem engineers' hain, soil formation+productivity badhate hain; mitti ko fine particles mein todkar soft banate hain (R **TRUE**). [Q12]

Agricultural Soils — Organic Matter, Sulphur Cycle, Salinization

- Organic matter soil ki water-holding capacity **badhata hai, kam NAHI karta** (Statement 1 galat); Soil **sulphur cycle mein role play karti hai** (plants sulphate absorb karte hain, decomposition sulphates wapas mitti mein laati hai — Statement 2 galat); **long-term irrigation se salinization ho sakta hai** (salts surface pe accumulate hote hain — Statement 3 **TRUE**). [Q13]

Edaphic Factor — Soil se Related

- **Edaphic** = soil ki physical/chemical composition se related abiotic factor hai (Climatic, Biotic, Topography se contrast). [Q14]

Geological Formations — Chronological Order

- Chronological order: **Sihawal (Lower Paleolithic) → Patpara (Middle Paleolithic) → Baghor (Upper Paleolithic) → Khetaurhi (Neolithic+Microlithic)** — yaani code 4,1,3,2. [Q15]

Geographical Cycle of Erosion — W.M. Davis

- **W.M. Davis (William Morris Davis)** ne geographical/geological cycle of erosion ki concept postulate ki — Walter Penck ne isse reject karke 'Morphological System' diya; Arthur Holmes-convection current theory; S.W. Wooldridge-London Basin structural evolution; Kober-Geosynclinal Orogen Theory. [Q16]